

Shamanth S Poojary

Embedded Firmware Engineer

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PROFESSIONAL SUMMARY

Embedded Firmware Engineer specializing in bare-metal and RTOS-based firmware development in Embedded C/C++ on ARM Cortex-M platforms, including STM32, ESP32, and Nordic nRF5. Develops low-level peripheral drivers (GPIO, I2C, SPI, UART), hardware abstraction layers, and LVGL-based embedded GUIs; owns board bring-up, firmware debugging, and hardware validation using JTAG/SWD, GDB, logic analyzers, and oscilloscopes. Currently developing TWS earbuds firmware on RTX5 RTOS with Bluetooth Classic and BLE (A2DP, HFP, SPP, GATT), with additional depth in Embedded Linux, Linux device drivers, and Yocto.

CORE TECHNICAL SKILLS

- **Programming Languages:** Embedded C, C, C++, Python, Shell Scripting, Data Structures & Algorithms (DSA)
- **Firmware & RTOS:** Bare-Metal Programming, RTOS (FreeRTOS, RTX5, Zephyr), ARM Cortex-M Firmware Development, Hardware Abstraction Layer (HAL), Peripheral Driver Development, Interrupt Handling (ISR), Board Bring-up, BSP (Board Support Package)
- **Microcontrollers & SoCs:** STM32 (ARM Cortex-M, STM32F1, STM32F4), ESP32 / ESP32-S3, Nordic nRF5 Series (Wireless SoC)
- **Interfaces & Protocols:** GPIO, I2C, SPI, UART, ADC, PWM, USB, EEPROM/Flash, Bluetooth Classic (A2DP, HFP, SPP, RFCOMM, SDP), BLE (GATT), BLE Mesh, TWS (True Wireless Stereo), Wi-Fi (802.11), TCP/IP, MQTT, 4-20mA Industrial Communication
- **Embedded Linux:** Linux System Programming, Linux Device Driver Development, Yocto Project (layers, recipes, minimal images), Kernel Cross-Compilation, System Calls, Multithreading, Inter-Process Communication (IPC), File Systems
- **Debugging, Testing & Validation:** GDB, JTAG/SWD, Logic Analyzer, Oscilloscope, Firmware Debugging, Hardware Validation, Unit Testing, Integration Testing, Test Framework Development, Code Review
- **Embedded GUI & Application Development:** LVGL, SquareLine Studio, Touchscreen Integration (GT911), Display Driver Integration, SPI/RGB Display Interfaces, Desktop Application Development (Qt, C++)
- **Tools:** Git, STM32CubeIDE, Simplicity Studio, PlatformIO, Segger Embedded Studio, VS Code, CMake, Makefiles, JIRA
- **Additional Platforms:** Raspberry Pi, BeagleV Ahead, RP2040, TI CC3200, Arduino (ATmega328P), 8051, PIC, MATLAB

PROFESSIONAL EXPERIENCE

Firmware Engineer, MIVI

June 2026 – Present

- Develop and debug TWS earbuds firmware in Embedded C on an RTX5 RTOS-based SoC platform, covering audio streaming, host connectivity and data communication.
- Implemented an embedded audio pipeline for TWS earbuds, integrating SPP-based audio streaming for Android and GATT-based audio streaming for iOS to support both mobile platforms.
- Debug and resolve firmware and connectivity issues across the Bluetooth and audio data paths, isolating faults between protocol handling, audio streaming and platform-specific behavior on Android and iOS.

Associate Software Engineer, SpanIdea Systems Pvt. Ltd.

June 2025 – May 2026

- Developed Linux device drivers and Linux system-programming applications on Raspberry Pi and BeagleV Ahead platforms, working across the kernel and user-space boundary.
- Developed bare-metal and FreeRTOS-based firmware for STM32, STM32-P107 and ESP32-LyraT Mini boards, exercising on-chip peripherals and validating behavior on target hardware.
- Built and customized embedded Linux distributions with the Yocto Project, authoring layers and recipes and generating minimal images for target hardware.
- Developed GNSS receiver configuration and validation software in Qt/C++ and C.

- Developed ESP32 firmware under FreeRTOS for touchscreen-based connected devices, implementing Wi-Fi provisioning and RTC time synchronization.
- Implemented low-level peripheral drivers and hardware abstraction layers (HAL) for STM32 microcontrollers, covering GPIO, I2C, SPI, UART, EEPROM and Flash memory.
- Designed and integrated LVGL-based embedded GUIs, including touchscreen interfacing and display driver development.
- Implemented BLE and BLE Mesh communication stacks and Flash drivers on Silicon Labs EFR32 wireless SoC platforms.
- Performed board bring-up, firmware debugging and hardware validation using logic analyzers and oscilloscopes to isolate peripheral and signal-level issues.
- Conducted unit testing and firmware debugging across driver and application layers to improve system reliability and reduce defects.

PROJECTS

GNSS Receiver Configuration Tool & Validation Testbench

SpanIdea | Jun 2025 – Jan 2026

- Built a Qt/C++ desktop tool that configures GNSS receivers over UART and renders live navigation data, including satellite tracking, signal strength and positional updates.
- Designed a C-based automated testbench for functional validation and performance evaluation of multiple GNSS receiver modules.
- Added feature enhancements and maintenance updates driven by the results coming out of validation runs.

Technologies: C, C++, Qt, UART, GNSS, VS Code, Segger Embedded Studio

Air Quality Monitoring Device – ESP32 Firmware & Embedded GUI

Zilliot | Feb 2025 – Jun 2025

- Developed C/C++ firmware for an ESP32-based touchscreen air quality monitoring device running under FreeRTOS.
- Integrated the SEN66 sensor over I2C and the APC1 sensor over UART, handling acquisition and processing of environmental parameters for on-device visualization.
- Designed an LVGL-based graphical interface using SquareLine Studio and implemented touchscreen event handling for user interaction.
- Implemented Wi-Fi connectivity for real-time RTC synchronization of the displayed measurements.

Technologies: ESP32, C/C++, FreeRTOS, I2C, UART, LVGL, SquareLine Studio, Wi-Fi, PlatformIO

Industrial Flow Meter Device – STM32 Firmware & Drivers

Zilliot | Aug 2024 – Dec 2024

- Developed STM32F1 firmware for a liquid flow measurement system, covering data acquisition, processing and real-time measurement display.
- Implemented low-level display and EEPROM/Flash drivers together with peripheral abstraction layers for GPIO, I2C, SPI and UART.
- Customized the display driver and created bitmap assets for measurement units and device branding.

Technologies: STM32F1, C/C++, FreeRTOS, I2C, SPI, UART, EEPROM/Flash, STM32CubeIDE

EDUCATION

N.M.A.M. Institute of Technology

Oct 2020 – May 2024

- **Bachelor of Engineering (B.E.), Electronics & Communication Engineering (ECE)**, Graduated May 2024
- **Relevant Coursework:** Embedded Systems, Microcontrollers, Real-Time Systems, Digital Electronics, Analog Electronics, Communication Systems, Internet of Things (IoT), Digital Signal Processing (DSP)